ANTIBACTERIAL AND ANTIOXIDANT ACTIVITIES OF HONEY FROM SOUTHWESTERN SLOVAKIA

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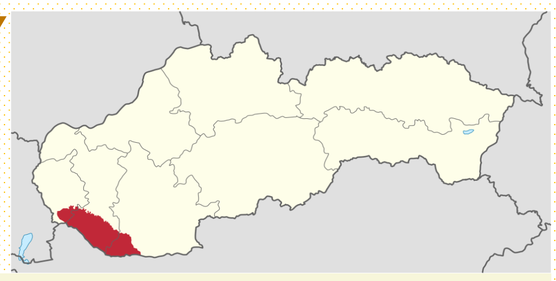


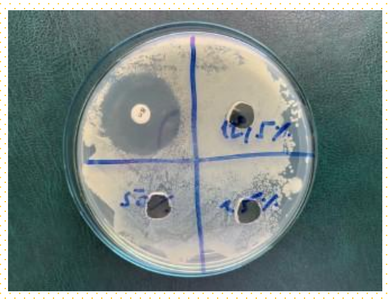
The aim of this study was to evaluate the physicochemical properties, antibacterial activity, and antioxidant potential of ten honey samples collected from the Žitný ostrov region in Southwestern Slovakia. Water activity and moisture content were determined, antibacterial activity was assessed against two Gram-positive bacteria (*Staphylococcus aureus* and *Enterococcus faecalis*) and one Gram-negative bacterium (*Escherichia coli*) at four honey concentrations (100%, 50%, 25%, and 12.5% v/v), and antioxidant activity together with total polyphenol content was analysed.

MATERIAL AND METHODOLOGY

The study was carried out on 10 honey samples collected from the Žitný ostrov region in Southwestern Slovakia. The antimicrobial activity of the honeys - well diffusion method. Serial dilution of honey - 100%, 50%, 25%, and 12.5%, 100 µL of inoculum in a concentration of 0.5 MFU - to a Petri dish with Mueller Hinton agar, Three bacteria - from the Czech collection of microorganisms of Masaryk University in Brno, each dilution of honey was added into the wells, the incubation at 37 °C for 24 h, the incubation zones of inhibition were measured. The antioxidant activity (DPPH method) and total polyphenols (spectrophotometrically) of honey samples were also evaluated.

RESULTS

Sample No.	Origin	Type of honey
1.	Baloň	Acacia honey
2.	Baloň	Phacelia honey
3.	Baloň	Flower honey
4.	Baloň	Linden honey
5.	Baloň	Rapeseed honey
6.	Jurová	Acacia honey
7.	Jurová	Spring honey
8.	Jurová	Rapeseed honey
9.	Jurová	Sunflower honey
10.	Jurová	Autumn honey



Physicochemical parameters of different types of honey

Sample No.	Type of honey	Water activity	Moisture content (%)
1.	Acacia	0.405	14.8
2.	Phacelia	0.403	15.5
3.	Flower	0.396	14.9
4.	Linden	0.399	14.2
5.	Rapeseed	0.401	15.4
6.	Acacia	0.450	18.3
7.	Spring	0.423	19.5
8.	Rapeseed	0.452	19.2
9.	Sunflower	0.445	19.6
10.	Autumn	0.447	18.3

Antibacterial effect of honey on growth of Gram-positive and Gram-negative bacteria

Sample No.	*Staphylococcus aureus*				*Enterococcus faecalis*			
	Honey concentrations (v/v) and diameter of inhibition zones (mm)							
	100%	50%	25%	12.5%	100%	50%	25%	12.5%
1.	10.5±0.57	8.75±0.96	5.72±0.49	4.5±0.57	10.75±0.96	7.5±0.41	4.75±0.96	1.00±0.82
2.	11.5±1.29	10.75±1.5	7.07±0.65	5.0±1.41	11.5±0.57	9.87±0.85	6.5±1.38	1.83±0.41
3.	10.25±0.5	6.00±1.26	4.12±0.86	1.25±0.5	9.00±0.00	6.83±0.41	4.44±0.45	1.00±0.00
4.	12.25±0.5	7.87±0.85	3.75±0.5	1.37±0.48	7.33±0.52	5.75±0.96	3.00±0.00	1.83±0.41
5.	13.6±0.58	7.67± 0.57	3.75±0.5	1.25±0.5	7.25±1.26	5.25±0.5	2.5± 0.57	NE
6.	10.25±0.5	7.00±0.81	4.00±0.82	1.00±0.00	7.75±0.5	4.0±0.00	1.25±0.5	1.00±0.00
7.	10.75±0.5	6.5±1.38	4.00±0.82	2.75±0.96	7.0±0.82	5.5±0.57	3.0±0.82	1.25±0.85
8.	10.5±0.57	9.25±0.5	4.5±1.29	3.25±0.5	6.25±0.64	1.0±0.00	NE	NE
9.	12.5±0.73	9.00±1.15	7.67±0.49	6.00±0.00	7.0±0.82	2.37±0.48	NE	NE
10.	10.25±0.5	5.75±0.96	4.5±0.57	2.27±0.49	7.00±0.00	4.33±0.57	NE	NE

Sample No.	Type of honey	*Escherichia coli*			
		Honey concentrations (v/v) and diameter of inhibition zones (mm)			
		100%	50%	25%	12.5%
1.	Acacia	12.67±0.53	8.33±0.53	5.00±1.00	4.33± 0.57
2.	Phacelia	11.67±0.57	6.33±0.57	5.00±1.00	3.00±1.00
3.	Flower	12.5±0.50	6.67± 0.58	5.60±1.51	4.33±0.57
4.	Linden	8.37±1.48	4.83±1.04	2.60±1.51	2.00±0.75
5.	Rapeseed	10.53±0.50	4.57±0.51	2.00±0.00	1.00±0.00
6.	Acacia	10.63±0.55	3.00±0,00	2.00±0,00	NE
7.	Spring	12.00±0.00	4.57±0.51	3.00±0.00	2.00±0.00
8.	Rapeseed	10.07±1.01	5.60±1.22	NE	NE
9.	Sunflower	10.55±0.51	5.83±1.04	1.00±0.00	NE
10.	Autumn	8.93±0.11	3.27±0.46	2.00±0.00	0.77±0.25

Antioxidant activity and total polyphenol content of tested honey

Sample No.	Type of honey	DPPH (mg TEAC/g)	Total polyphenols (mg GAE/g)
1.	Acacia	0.86 ±0.02	0.53 ±0.01
2.	Phacelia	0.91 ±0.12	0.44 ±0.02
3.	Flower	0.88 ±0.02	0.39 ±0.03
4.	Linden	0.82 ±0.03	8.97 ±0.01
5.	Rapeseed	0.89 ±0.01	5.32 ±0.04
6.	Acacia	0.79 ±0.04	0.58 ±0.02
7.	Spring	0.86 ±0.01	0.41 ±0.02
8.	Rapeseed	0.91 ±0.02	0.31 ±0.03
9.	Sunflower	0.95 ±0.03	0.79 ±0.01
10.	Autumn	1.03 ±0.02	0.74 ±0.02

CONCLUSION

The present study confirmed that all analyzed honey samples met the physicochemical quality requirements, with moisture content within EU limits and water activity values below 0.60, indicating good microbiological stability..The results demonstrated a clear concentration-dependent antibacterial effect of honey against *Staphylococcus aureus*, *Enterococcus faecalis*, and *Escherichia coli*, with notable variability among different honey types. Phacelia honey exhibited the most consistent antibacterial activity across all tested concentrations, while acacia and flower honeys showed strong effects particularly against *E. coli*. The reduced sensitivity of *E. faecalis* compared to *S. aureus* highlights differences in bacterial susceptibility to honey. Antioxidant analyses revealed that darker honeys, particularly autumn honey, possessed higher radical scavenging activity, whereas total polyphenol content did not always correlate directly with antioxidant capacity, as demonstrated by linden honey. Overall, the findings emphasize the multifactorial nature of honey's biological activity and confirm that botanical origin, geographical location, and concentration play a crucial role in determining both antibacterial and antioxidant properties.

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